

Regional Breast Pathology Service

Integrated Diagnostic Pathology Report

Accession:	RBPS-2026-004115	Patient ID:	PID-29103
Report date:	2026-04-29	Age / Sex:	52 / F

Clinical history / indication

52-year-old postmenopausal woman with a screening-detected 2.1 cm spiculated mass in the upper outer quadrant of the left breast. Core needle biopsy showed invasive carcinoma. Underwent partial mastectomy with sentinel lymph node biopsy. No personal history of breast cancer. Mother diagnosed with breast cancer at age 68; otherwise no significant family history.

Specimen

Left breast partial mastectomy with sentinel lymph node biopsy (3 nodes).

Macroscopic findings

Fibrofatty tissue 7.5 x 6.0 x 3.1 cm containing a firm, spiculated, gray-white tumor measuring 2.1 x 1.9 x 1.6 cm. Closest margin: 4 mm (anterior). Three sentinel lymph nodes, largest 1.4 cm.

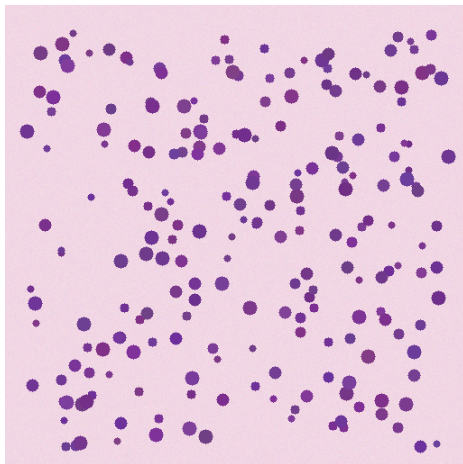
Microscopic / histopathology

Sections demonstrate invasive carcinoma of no special type composed of nests and cords of moderately atypical epithelial cells infiltrating fibrous stroma. Nottingham grade is 2 (tubule formation 3, nuclear pleomorphism 2, mitoses 2; total score 7/9). No lymphovascular invasion is identified. Margins are negative; the closest margin is 4 mm anteriorly. Associated DCIS, intermediate grade, solid pattern, is present and accounts for approximately 10% of the tumor. All three sentinel lymph nodes are negative for metastatic carcinoma (0/3).

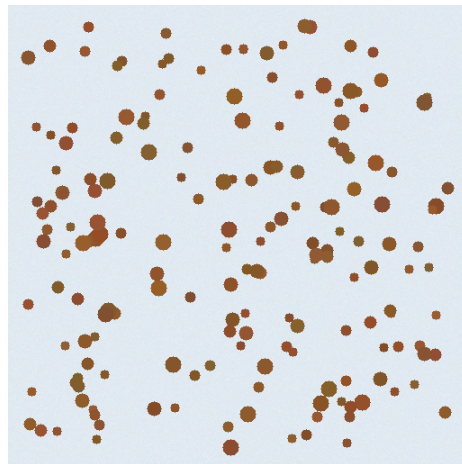
Immunohistochemistry

Marker	Result
Estrogen receptor (ER)	Positive, 95% strong nuclear (Allred 8/8)
Progesterone receptor (PR)	Positive, 40% strong nuclear (Allred 7/8)
HER2 (4B5)	1+ (negative); no further FISH testing indicated
Ki-67 (MIB-1)	Approximately 18%
E-cadherin	Positive (retained membranous)
AE1/AE3	Positive

Representative images



H&E; ×20 — invasive ductal carcinoma



ER IHC ×20

Molecular findings

Single-nucleotide variants / indels

Gene	HGVSc	HGVSp	VAF	Classification
PIK3CA	c.3140A>G	p.H1047R	0.32	Pathogenic / Actionable

Copy-number alterations

Gene	Copy number	Kind
ERBB2	2	no amplification

MSI status	MSS
TMB (mut/Mb)	3.6
Germline screen	BRCA1 / BRCA2 / PALB2: no pathogenic variants identified

Pathologist comments

Invasive carcinoma of no special type, Nottingham grade 2 (3+2+2=7/9), 2.1 cm, ER+/PR+/HER2-. PIK3CA H1047R detected (somatic, VAF ~32%) — actionable for alpelisib + endocrine therapy consideration in advanced/metastatic settings per current NCCN. Margins negative (closest 4 mm anterior). 0/3 sentinel nodes positive. Recommend multidisciplinary discussion of adjuvant endocrine therapy +/- chemotherapy based on recurrence-risk assays.

Electronically signed: **Reporting Pathologist, MD, PhD**
2026-04-29